

ORDERED T_N -CONFIGURATIONS IN CONSTITUTIVE SETS FOR 2×2 HYPERBOLIC SYSTEMS WITH ONE ENTROPY: A COUNTEREXAMPLE TO A WEAK FORMULATION AND STRUCTURAL OBSTRUCTIONS

ABSTRACT. Let $V \subset \mathbb{R}^2$ be a nonempty convex open set, let $F: V \rightarrow \mathbb{R}^2$ be the flux of a strictly hyperbolic and genuinely nonlinear system, and suppose that the system admits a strictly convex entropy–entropy-flux pair (η, q) . The associated constitutive surface is

$$K_{F,\eta,q} = \left\{ \begin{pmatrix} u_1 & -F_1(U) \\ u_2 & -F_2(U) \\ \eta(U) & -q(U) \end{pmatrix} : U = (u_1, u_2) \in V \right\} \subset \mathbb{R}^{3 \times 2}.$$

This note records a careful resolution of several logically distinct questions concerning ordered T_N -configurations in $K_{F,\eta,q}$. First, if the definition of an ordered T_N -configuration does not require the outer states to be pairwise rank-two connected, then the proposed universal nonexistence statement is false. We give an explicit smooth strictly hyperbolic, genuinely nonlinear system with a quadratic strictly convex entropy whose constitutive set contains a rank-one pair; that pair generates a degenerate ordered T_N for every even $N \geq 4$.

We then turn to the genuinely harder rank-one-free problem. We prove that for a cubic potential flux $F = \nabla \Phi$ with quadratic entropy, strict hyperbolicity and genuine nonlinearity rule out all rank-one connections. We also exhibit exact nondegenerate ordered T_4 and T_6 configurations whose first columns lie on a strictly convex entropy graph and show explicitly how each finite configuration can be interpolated by a single entropy-compatible polynomial potential system. The simplest separable realization of the displayed T_4 necessarily loses genuine nonlinearity, while no global hyperbolicity or rank-one-freeness claim is made for the displayed T_6 interpolant. For arbitrary N , we derive a positive-weight determinant identity for every 2×2 minor of an ordered T_N . This identity rules out configurations supported on a one-dimensional constitutive curve in a large class of separable, rank-one-free systems. Finally, we prove a local obstruction: every regular one-scale family of T_N -configurations collapsing to a strictly hyperbolic state is collinear at first order. Consequently, increasing from $N = 4$ to $N = 6$ does not by itself remove the principal obstructions. The construction of a nondegenerate T_N inside an entirely rank-one-free constitutive surface satisfying all of the stated differential conditions is not completed here; the precise remaining problem is isolated.

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1. INTRODUCTION

Consider the one-dimensional system of conservation laws

$$(1.1) \quad U_t + F(U)_x = 0, \quad U: \mathbb{R} \times [0, \infty) \rightarrow V \subset \mathbb{R}^2,$$

where V is a nonempty convex open set and $F = (F_1, F_2): V \rightarrow \mathbb{R}^2$ is smooth. Suppose that there is an entropy–entropy-flux pair (η, q) satisfying

$$(1.2) \quad \nabla q(U) = \nabla \eta(U) DF(U), \quad U \in V,$$

and that $D^2\eta(U)$ is positive definite throughout V . We assume that $DF(U)$ has two distinct real eigenvalues

$$(1.3) \quad \lambda_1(U) < \lambda_2(U)$$

with corresponding right eigenvectors $r_1(U), r_2(U)$, and that both characteristic fields are genuinely nonlinear:

$$(1.4) \quad \nabla \lambda_k(U) \cdot r_k(U) \neq 0, \quad k = 1, 2, \quad U \in V.$$

The system together with the entropy equality can be encoded by the constitutive map

$$(1.5) \quad G(U) := \begin{pmatrix} u_1 & -F_1(U) \\ u_2 & -F_2(U) \\ \eta(U) & -q(U) \end{pmatrix}, \quad U = (u_1, u_2),$$

and the constitutive set

$$(1.6) \quad K_{F,\eta,q} := G(V) \subset \mathbb{M}, \quad \mathbb{M} := \mathbb{R}^{3 \times 2}.$$

For an integer $N \geq 1$, we distinguish the single-matrix constitutive set from its N -fold product by writing

$$(1.7) \quad \mathcal{K}_{F,\eta,q}^{(N)} := K_{F,\eta,q}^N \subset \mathbb{M}^N.$$

The rank-one geometry of sets such as (1.6) is central to the differential-inclusion approach to conservation laws initiated in [KMS03]. For the p -system, nonexistence of T_4 -configurations was proved by Lorent and Peng [LP20], and nonexistence of T_5 -configurations was proved by Johansson and Tione [JT24]. For broad classes of general 2×2 systems satisfying additional global assumptions on the shock curves, T_4 -configurations are excluded in [KS24]. Peng and Vuolo [PV24] exclude two-dimensional T_N -configurations and a specified class of three-dimensional

T_N -configurations for the p -system, while explicitly leaving the unrestricted general- N problem open in that setting. Abstract matrix sets without rank-one connections can nevertheless carry nontrivial laminate geometry; see, for example, [Iqb00, Szé05]. The issue addressed here is the additional rigidity imposed by the constitutive form (1.5).

The starting question was whether the local assumptions (1.2)–(1.4) alone imply that $K_{F,\eta,q}$ contains no ordered T_N for any N . There are two versions of this question, and they must not be conflated.

- (i) In a *weak algebraic definition*, one asks only for a closed rank-one polygon and outer parameters $\kappa_i > 1$. Distinctness of the outer matrices and the condition
- $$(1.8) \quad \text{rank}(X_i - X_j) = 2, \quad i \neq j,$$

are not imposed.

- (ii) In the standard nondegenerate definition used in the differential-inclusion literature, the outer matrices have no rank-one connections; in our 3×2 setting, this is precisely (1.8).

Under the weak definition, the proposed universal nonexistence assertion is false. Indeed, one rank-one pair produces a degenerate ordered T_N for every even N , and in section 3 we construct a system satisfying all of the local differential assumptions whose constitutive set contains such a pair.

The standard rank-one-free problem is much subtler. The results of this note do not produce the requested nondegenerate example. Instead, they isolate several rigorous facts that any proposed T_6 , or more generally any proposed T_N , must satisfy:

- cubic potential systems provide a clean rank-one-free class;
- strict entropy convexity at finitely many states is compatible with exact nondegenerate T_4 and T_6 configurations;
- every ordered T_N satisfies a positive-weight identity for each 2×2 minor;
- one-dimensional constructions in a broad separable class are impossible for every N ;
- regular small-amplitude configurations are necessarily collinear at first order.

The exact T_6 constructed in section 6 confirms that increasing the number of states creates substantial finite-dimensional algebraic freedom. It does not, however, avoid the determinant or local asymptotic obstructions, and it does not by itself produce a globally rank-one-free hyperbolic constitutive surface.

2. PRELIMINARIES

2.1. Entropy compatibility and symmetrization. We use row-vector notation for gradients. Differentiating (1.2) gives the familiar symmetrization property.

Lemma 2.1 (Entropy symmetrizer). *Assume $F, \eta, q \in C^2(V)$ and (1.2). Then*

$$(2.1) \quad D^2\eta(U) DF(U) \text{ is symmetric for every } U \in V.$$

Proof. Differentiate (1.2). In coordinates,

$$\partial_j q = \sum_{\alpha=1}^2 \partial_\alpha \eta \partial_j F_\alpha.$$

Therefore

$$\partial_k \partial_j q = \sum_{\alpha=1}^2 \partial_k \partial_\alpha \eta \partial_j F_\alpha + \sum_{\alpha=1}^2 \partial_\alpha \eta \partial_k \partial_j F_\alpha.$$

Equivalently,

$$(2.2) \quad D^2q = D^2\eta DF + \sum_{\alpha=1}^2 \partial_\alpha \eta D^2F_\alpha.$$

Both D^2q and each D^2F_α are symmetric. Hence the first term on the right-hand side of (2.2) is symmetric, which is (2.1). $\square$

A particularly useful subclass is obtained from a potential.

Lemma 2.2 (Quadratic entropy and potential flux). *Let $V \subset \mathbb{R}^2$ be convex and let*

$$(2.3) \quad \eta(U) = \frac{1}{2} |U|^2.$$

If $F \in C^2(V; \mathbb{R}^2)$ admits an entropy flux q satisfying (1.2), then DF is symmetric. Consequently, there exists $\Phi \in C^3(V)$ such that

$$(2.4) \quad F = \nabla \Phi.$$

Conversely, if (2.4) holds, then

$$(2.5) \quad q(U) = U \cdot \nabla \Phi(U) - \Phi(U)$$

is an associated entropy flux, up to an additive constant.

Proof. For (2.3), the compatibility relation becomes

$$\partial_j q(U) = \sum_{\alpha=1}^2 u_\alpha \partial_j F_\alpha(U).$$

Taking the difference of the mixed derivatives gives

$$0 = \partial_k \partial_j q - \partial_j \partial_k q = \partial_j F_k - \partial_k F_j.$$

Thus $DF = DF^\top$. Since V is convex and hence simply connected, F is a gradient. Conversely, differentiating (2.5) gives

$$\nabla q(U) = U^\top D^2 \Phi(U) = \nabla \eta(U) DF(U).$$

$\square$

2.2. Rank-one connections.

Definition 2.3. Two distinct matrices $A, B \in \mathbb{M}$ are *rank-one connected* if

$$\text{rank}(B - A) = 1.$$

The set $K_{F,\eta,q}$ is *rank-one-free* if it contains no such pair.

For constitutive matrices, rank-one connections are exactly Rankine–Hugoniot discontinuities with zero entropy production.

Lemma 2.4 (Rank-one criterion). *Let $U_-, U_+ \in V$ be distinct. Then*

$$(2.6) \quad \text{rank}(G(U_+) - G(U_-)) = 1$$

if and only if there exists $\sigma \in \mathbb{R}$ such that

$$(2.7) \quad F(U_+) - F(U_-) = \sigma(U_+ - U_-),$$

$$(2.8) \quad q(U_+) - q(U_-) = \sigma(\eta(U_+) - \eta(U_-)).$$

Proof. The first column of $G(U_+) - G(U_-)$ is

$$\begin{pmatrix} U_+ - U_- \\ \eta(U_+) - \eta(U_-) \end{pmatrix},$$

which is nonzero because $U_+ \neq U_-$. Hence (2.6) holds exactly when the second column is a scalar multiple of the first. Because of the minus signs in (1.5), that scalar can be written as $-\sigma$, and the two block identities are precisely (2.7) and (2.8). $\square$

Remark 2.5 (Sign convention). Some references define the constitutive map with second column $(F_1, F_2, q)^\top$ rather than $(-F_1, -F_2, -q)^\top$. The two conventions are related by right multiplication by $\text{diag}(1, -1)$. This invertible linear transformation preserves ranks, rank-one connections, and ordered T_N -configurations.

2.3. Weak and nondegenerate ordered T_N -configurations. Set

$$(2.9) \quad \mathcal{R}_1 := \{C \in \mathbb{M} : \text{rank } C = 1\}.$$

Thus $\mathcal{R}_1$ is the nonzero rank-one manifold in $\mathbb{M}$.

Definition 2.6 (Weak ordered T_N -configuration). Fix $N \geq 4$. An ordered tuple

$$\mathbf{X} = (X_1, \dots, X_N) \in \mathbb{M}^N$$

is a *weak ordered T_N -configuration* if there exist

$$(2.10) \quad P \in \mathbb{M}, \quad C_i \in \mathcal{R}_1, \quad \kappa_i > 1, \quad i = 1, \dots, N,$$

such that

$$(2.11) \quad \sum_{i=1}^N C_i = 0$$

and

$$(2.12) \quad X_i = P + \sum_{j=1}^{i-1} C_j + \kappa_i C_i, \quad i = 1, \dots, N.$$

The collection

$$\theta = (P, C_1, \dots, C_N, \kappa_1, \dots, \kappa_N)$$

is called an ordered T_N -parameterization of $\mathbf{X}$.

Definition 2.7 (Nondegenerate ordered T_N -configuration). A weak ordered T_N -configuration is *nondegenerate* if

$$(2.13) \quad \text{rank}(X_i - X_j) = 2, \quad i \neq j.$$

In particular, the X_i are distinct and no two outer states are rank-one connected.

Introduce the polygon vertices

$$(2.14) \quad P_i := P + \sum_{j=1}^{i-1} C_j, \quad i = 1, \dots, N+1.$$

Then (2.11) gives $P_{N+1} = P_1 = P$, and

$$(2.15) \quad P_{i+1} = P_i + C_i, \quad X_i = P_i + \kappa_i C_i.$$

Equivalently, with

$$(2.16) \quad \alpha_i := \frac{1}{\kappa_i}, \quad \beta_i := 1 - \frac{1}{\kappa_i},$$

we have $0 < \alpha_i, \beta_i < 1$ and

$$(2.17) \quad P_{i+1} = \beta_i P_i + \alpha_i X_i.$$

This elementary convex recursion will be used repeatedly.

Remark 2.8 (Affine invariance). Translations in matrix space and maps of the form

$$X \mapsto LXR, \quad L \in GL(3), \quad R \in GL(2),$$

preserve rank-one matrices, pairwise ranks, and both definitions of ordered T_N -configuration. Indeed, the same transformation can be applied to P , every C_i , and every X_i in (2.12).

3. A COUNTEREXAMPLE UNDER THE WEAK DEFINITION

3.1. Every rank-one pair generates weak T_N -configurations. The following observation explains why the pairwise rank-two condition is not optional if one wants a meaningful universal nonexistence statement.

Proposition 3.1 (Rank-one pair $\implies$ weak T_{2m}). *Let $A, B \in \mathbb{M}$ be distinct and suppose that*

$$R := B - A$$

has rank one. Then, for every even integer $N = 2m \geq 4$, the alternating tuple

$$(3.1) \quad (A, B, A, B, \dots, A, B) \in \mathbb{M}^N$$

is a weak ordered T_N -configuration.

Proof. Set

$$(3.2) \quad P := A + \frac{2}{3}R, \quad \kappa_i := 2,$$

and define

$$(3.3) \quad C_i := \begin{cases} -\frac{1}{3}R, & i \text{ odd}, \\ \frac{1}{3}R, & i \text{ even}. \end{cases}$$

Each C_i is nonzero and rank one. Since N is even, the positive and negative arms cancel, and therefore

$$\sum_{i=1}^N C_i = 0.$$

If i is odd, the partial sum before the i th arm vanishes:

$$\sum_{j < i} C_j = 0.$$

Consequently,

$$P + \sum_{j < i} C_j + 2C_i = A + \frac{2}{3}R - \frac{2}{3}R = A.$$

If i is even, then

$$\sum_{j < i} C_j = -\frac{1}{3}R,$$

and hence

$$P + \sum_{j < i} C_j + 2C_i = A + \frac{2}{3}R - \frac{1}{3}R + \frac{2}{3}R = A + R = B.$$

Thus (2.12) gives exactly the alternating tuple (3.1). □

3.2. An explicit entropy system with a rank-one connection. We next construct an explicit system satisfying strict hyperbolicity, genuine nonlinearity, and strict entropy convexity on a convex open state space, while its constitutive set contains a rank-one pair.

Write $U = (x, y)$ and define the polynomial potential

$$(3.4) \quad \begin{aligned} \Phi(x, y) := & \frac{3}{4}x^4 + \frac{7}{6}(x^3 - x)y + \left(1 - \frac{5}{2}x\right)y^2 \\ & - \left(\frac{10}{3} + \frac{25}{6}x\right)y^3. \end{aligned}$$

Set

$$(3.5) \quad F := \nabla \Phi, \quad \eta(U) := \frac{1}{2}|U|^2, \quad q(U) := U \cdot \nabla \Phi(U) - \Phi(U).$$

By [lemma 2.2](#), (η, q) is an entropy–entropy-flux pair, and

$$(3.6) \quad D^2\eta(U) = I_2$$

is positive definite everywhere.

The first derivatives of Φ are

$$(3.7) \quad F_1(x, y) = 3x^3 + \left(\frac{7}{2}x^2 - \frac{7}{6}\right)y - \frac{5}{2}y^2 - \frac{25}{6}y^3,$$

$$(3.8) \quad F_2(x, y) = \frac{7}{6}(x^3 - x) + (2 - 5x)y - \left(10 + \frac{25}{2}x\right)y^2.$$

Along the segment

$$(3.9) \quad I := [-1, 1] \times \{0\},$$

we have

$$(3.10) \quad F(x, 0) = \begin{pmatrix} 3x^3 \\ \frac{7}{6}(x^3 - x) \end{pmatrix}, \quad q(x, 0) = \frac{9}{4}x^4.$$

Let

$$(3.11) \quad U_- := (-1, 0), \quad U_+ := (1, 0).$$

Then

$$(3.12) \quad F(U_-) = (-3, 0), \quad F(U_+) = (3, 0),$$

while

$$(3.13) \quad \eta(U_-) = \eta(U_+) = \frac{1}{2}, \quad q(U_-) = q(U_+) = \frac{9}{4}.$$

Consequently,

$$(3.14) \quad G(U_-) = \begin{pmatrix} -1 & 3 \\ 0 & 0 \\ \frac{1}{2} & -\frac{9}{4} \end{pmatrix}, \quad G(U_+) = \begin{pmatrix} 1 & -3 \\ 0 & 0 \\ \frac{1}{2} & -\frac{9}{4} \end{pmatrix},$$

and

$$(3.15) \quad G(U_+) - G(U_-) = \begin{pmatrix} 2 & -6 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & -3 \end{pmatrix}.$$

Thus the two constitutive matrices are rank-one connected. Equivalently, (2.7)–(2.8) hold with shock speed $\sigma = 3$.

It remains to verify the differential hypotheses on a convex neighborhood of I .

3.2.1. *Strict hyperbolicity on the segment.* Along I , the Jacobian is the symmetric Hessian

$$(3.16) \quad DF(x, 0) = D^2\Phi(x, 0) = \begin{pmatrix} 9x^2 & \frac{7}{6}(3x^2 - 1) \\ \frac{7}{6}(3x^2 - 1) & 2 - 5x \end{pmatrix}.$$

A real symmetric 2×2 matrix has a repeated eigenvalue exactly when its off-diagonal entry vanishes and its two diagonal entries agree. Applied to (3.16), this would require

$$(3.17) \quad 3x^2 - 1 = 0$$

and

$$(3.18) \quad 9x^2 = 2 - 5x.$$

Equation (3.17) gives $9x^2 = 3$, so (3.18) becomes

$$3 = 2 - 5x, \quad \text{that is,} \quad x = -\frac{1}{5}.$$

This is incompatible with $x^2 = 1/3$. Hence the two eigenvalues are distinct for every $x \in [-1, 1]$.

3.2.2. *Genuine nonlinearity on the segment.* We first recall the standard eigenvalue derivative formula for a symmetric potential system.

Lemma 3.2 (Directional derivative of an eigenvalue). *Let $\Phi \in C^3(V)$, let $A(U) := D^2\Phi(U)$, and suppose that $\lambda_k(U)$ is a simple eigenvalue of $A(U)$ with a unit eigenvector $r_k(U)$. Then*

$$(3.19) \quad \nabla \lambda_k(U) \cdot Z = D^3\Phi(U)[Z, r_k(U), r_k(U)]$$

for every $Z \in \mathbb{R}^2$. In particular,

$$(3.20) \quad \nabla \lambda_k(U) \cdot r_k(U) = D^3\Phi(U)[r_k(U), r_k(U), r_k(U)].$$

Proof. Differentiate the identity

$$A(U)r_k(U) = \lambda_k(U)r_k(U)$$

in the direction Z and take the scalar product with $r_k(U)$. Since $A(U)$ is symmetric and $|r_k(U)| = 1$, the terms containing the derivative of r_k cancel, leaving

$$D\lambda_k(U)[Z] = r_k(U)^\top (DA(U)[Z])r_k(U),$$

which is (3.19). □

Along I , the independent components of the third derivative of (3.4) are

$$(3.21) \quad \Phi_{xxx} = 18x, \quad \Phi_{xyx} = 7x, \quad \Phi_{xyy} = -5, \quad \Phi_{yyy} = -(20 + 25x).$$

Let $r = (a, b)$ be an eigenvector of (3.16).

Suppose first that $a \neq 0$ and put

$$(3.22) \quad t := \frac{b}{a}.$$

The eigenvector condition for $(1, t)$ is equivalent to

$$(3.23) \quad Q(x, t) = 0,$$

where

$$(3.24) \quad Q(x, t) := \frac{7}{6}(3x^2 - 1)t^2 + (9x^2 + 5x - 2)t - \frac{7}{6}(3x^2 - 1).$$

Indeed, if

$$\begin{pmatrix} A & B \\ B & C \end{pmatrix} \begin{pmatrix} 1 \\ t \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ t \end{pmatrix},$$

then elimination of λ gives $Bt^2 + (A - C)t - B = 0$, which is exactly (3.24).

By (3.21),

$$(3.25) \quad \begin{aligned} D^3\Phi(x, 0)[(1, t), (1, t), (1, t)] &:= H(x, t) \\ &= 18x + 21xt - 15t^2 - (20 + 25x)t^3. \end{aligned}$$

Since $r = a(1, t)$, trilinearity gives $D^3\Phi(x, 0)[r, r, r] = a^3H(x, t)$; because $a \neq 0$, this quantity vanishes if and only if $H(x, t) = 0$. Thus genuine nonlinearity could fail for an eigenvector with $a \neq 0$ only if the two polynomials $Q(x, \cdot)$ and $H(x, \cdot)$ have a common real root. Their resultant with respect to t is

$$(3.26) \quad \text{Res}_t(Q, H) = \frac{1}{216} \mathcal{P}(x),$$

where

$$(3.27) \quad \begin{aligned} \mathcal{P}(x) &:= 129007296x^8 + 235355760x^7 + 80940141x^6 \\ &\quad - 64110420x^5 - 27496701x^4 + 10403280x^3 \\ &\quad + 1618495x^2 - 1203500x + 236425. \end{aligned}$$

We now verify that $\mathcal{P}$ has no zero on $[-1, 1]$. Let

$$(3.28) \quad S_0 := \mathcal{P}, \quad S_1 := \mathcal{P}', \quad S_{j+1} := -\text{rem}(S_{j-1}, S_j)$$

be the Sturm chain, with arbitrary positive rescalings of its members allowed. Exact rational arithmetic gives the endpoint signs

$$(3.29) \quad \begin{array}{c|cccccccccc} & S_0 & S_1 & S_2 & S_3 & S_4 & S_5 & S_6 & S_7 & S_8 \\ \hline x = -1 & + & - & + & + & - & - & - & + & + \\ x = 1 & + & + & + & - & - & + & - & + & + \end{array}$$

with no zero entries. Both rows have four sign changes. Sturm's theorem therefore yields

$$(3.30) \quad \#\{x \in (-1, 1) : \mathcal{P}(x) = 0\} = 4 - 4 = 0.$$

Since $\mathcal{P}(0) = 236425 > 0$, it follows that

$$(3.31) \quad \mathcal{P}(x) > 0, \quad -1 \leq x \leq 1.$$

The resultant in (3.26) is therefore nonzero on the entire segment, so (3.23) and (3.25) cannot hold simultaneously.

It remains to consider an eigenvector with $a = 0$. Such an eigenvector is parallel to $e_2 = (0, 1)$, which can occur only when the off-diagonal entry in (3.16) vanishes. Hence $x^2 = 1/3$. At such a point,

$$(3.32) \quad D^3\Phi(x, 0)[e_2, e_2, e_2] = -(20 + 25x).$$

The right-hand side could vanish only at $x = -4/5$, which does not satisfy $x^2 = 1/3$. Therefore (3.20) is nonzero for both eigenfields at every point of I .

3.2.3. *Extension to a convex open state space.* Let

$$(3.33) \quad \gamma(U) := \lambda_2(U) - \lambda_1(U)$$

be the spectral gap. On the compact segment I we have

$$\gamma(U) > 0.$$

Moreover, after locally choosing unit eigenvectors and taking absolute values, the genuine-nonlinearity quantities satisfy

$$|\nabla \lambda_k(U) \cdot r_k(U)| > 0, \quad U \in I, \quad k = 1, 2.$$

These quantities are continuous on the strictly hyperbolic region. Hence there is an open neighborhood W of I on which the spectral gap remains positive and both genuine-nonlinearity quantities remain nonzero. Choose $\rho > 0$ sufficiently small that

$$(3.34) \quad V := I + B_\rho(0) = \left\{ U \in \mathbb{R}^2 : \text{dist}(U, I) < \rho \right\} \subset W.$$

Because I and $B_\rho(0)$ are convex, their Minkowski sum V is open and convex. The system is strictly hyperbolic and genuinely nonlinear throughout V , and (3.6) holds there.

Combining the preceding calculations gives the promised counterexample.

Theorem 3.3 (Failure of universal nonexistence under the weak definition). *There exist a nonempty convex open set $V \subset \mathbb{R}^2$, a smooth flux $F: V \rightarrow \mathbb{R}^2$, and a smooth entropy–entropy-flux pair (η, q) such that*

- (i) $D^2\eta(U) > 0$ for every $U \in V$;
- (ii) the system is strictly hyperbolic on V ;
- (iii) both characteristic fields are genuinely nonlinear on V ;
- (iv) $K_{F,\eta,q}$ contains a rank-one pair.

Consequently, for every even $N \geq 4$, the product $\mathcal{K}_{F,\eta,q}^{(N)}$ contains a weak ordered T_N -configuration.

Proof. Take (3.4)–(3.5) on the convex tube (3.34). The preceding subsections prove (i)–(iv). Apply proposition 3.1 to the two matrices in (3.14). $\square$

Remark 3.4. The configurations produced by theorem 3.3 repeat the two outer matrices and therefore fail (2.13). They disprove only the assertion based on definition 2.6. They do not answer the standard rank-one-free problem.

4. A RANK-ONE-FREE CLASS: CUBIC POTENTIAL SYSTEMS

The preceding counterexample shows that the local hyperbolic hypotheses do not by themselves rule out rank-one connections. There is, however, a large and very transparent subclass in which genuine nonlinearity does rule them out.

Proposition 4.1 (Cubic potential lemma). *Let $V \subset \mathbb{R}^2$ be convex, let $\Phi: V \rightarrow \mathbb{R}$ be the restriction of a polynomial of degree at most three, and define*

$$(4.1) \quad F = \nabla \Phi, \quad \eta(U) = \frac{1}{2} |U|^2, \quad q(U) = U \cdot \nabla \Phi(U) - \Phi(U).$$

Assume that $D^2\Phi(U)$ has two distinct eigenvalues throughout V and that

$$(4.2) \quad D^3\Phi[r_k(U), r_k(U), r_k(U)] \neq 0, \quad k = 1, 2, \quad U \in V,$$

for corresponding nonzero eigenvectors. Then $K_{F,\eta,q}$ is rank-one-free.

Proof. Suppose, toward a contradiction, that $U, W \in V$, $U \neq W$, are rank-one connected. Set

$$(4.3) \quad d := W - U, \quad m := \frac{U + W}{2}.$$

By [lemma 2.4](#), there exists $\sigma \in \mathbb{R}$ such that

$$(4.4) \quad \nabla \Phi(W) - \nabla \Phi(U) = \sigma d.$$

Because Φ is cubic, $D^2\Phi$ is affine. Therefore its average along the segment is its value at the midpoint:

$$(4.5) \quad \begin{aligned} \nabla \Phi(W) - \nabla \Phi(U) &= \int_0^1 D^2\Phi(U + td) d \, dt \\ &= D^2\Phi(m) d. \end{aligned}$$

Combining (4.4) and (4.5), we obtain

$$(4.6) \quad D^2\Phi(m) d = \sigma d.$$

Thus d is an eigenvector of $D^2\Phi(m)$.

The entropy equality in [lemma 2.4](#) is

$$(4.7) \quad q(W) - q(U) = \sigma(\eta(W) - \eta(U)).$$

Using $W = m + d/2$, $U = m - d/2$, and (4.4), we compute

$$(4.8) \quad \begin{aligned} q(W) - q(U) &= [W \cdot \nabla \Phi(W) - U \cdot \nabla \Phi(U)] - [\Phi(W) - \Phi(U)] \\ &= m \cdot [\nabla \Phi(W) - \nabla \Phi(U)] + \frac{1}{2} d \cdot [\nabla \Phi(W) + \nabla \Phi(U)] - [\Phi(W) - \Phi(U)] \\ &= \sigma m \cdot d + \frac{1}{2} d \cdot [\nabla \Phi(W) + \nabla \Phi(U)] - [\Phi(W) - \Phi(U)]. \end{aligned}$$

Also,

$$(4.9) \quad \eta(W) - \eta(U) = m \cdot d.$$

Substitution into (4.7) yields

$$(4.10) \quad \frac{1}{2} d \cdot [\nabla \Phi(W) + \nabla \Phi(U)] - [\Phi(W) - \Phi(U)] = 0.$$

Define the one-variable cubic polynomial

$$(4.11) \quad g(t) := \Phi(U + td), \quad 0 \leq t \leq 1.$$

Then

$$g'(0) = \nabla \Phi(U) \cdot d, \quad g'(1) = \nabla \Phi(W) \cdot d.$$

The exact trapezoidal-rule identity for a quadratic polynomial g' is

$$(4.12) \quad \frac{g'(0) + g'(1)}{2} - [g(1) - g(0)] = \frac{1}{12} g'''.$$

For completeness, this follows by checking the identity on the basis $1, t, t^2, t^3$, or by writing $g(t) = a + bt + ct^2 + dt^3$ and observing that both sides equal $d/2$. Combining (4.10) and (4.12) gives

$$(4.13) \quad D^3\Phi[d, d, d] = g''' = 0.$$

But (4.6) says that d is a nonzero eigenvector at m , and (4.2) says that the left-hand side of (4.13) is nonzero. This contradiction proves the claim. $\square$

Corollary 4.2 (An explicit rank-one-free system). *Let*

$$(4.14) \quad V := \{(x, y) \in \mathbb{R}^2 : x < y\}$$

and define

$$(4.15) \quad \Phi(x, y) := \frac{1}{6}(x^3 + y^3), \quad F(x, y) = \left(\frac{x^2}{2}, \frac{y^2}{2}\right),$$

$$(4.16) \quad \eta(x, y) = \frac{1}{2}(x^2 + y^2), \quad q(x, y) = \frac{1}{3}(x^3 + y^3).$$

Then V is nonempty, convex, and open; the system is strictly hyperbolic and genuinely nonlinear on V ; $D^2\eta = I_2$; and $K_{F,\eta,q}$ has no rank-one connections.

Proof. We have

$$DF(x, y) = \text{diag}(x, y).$$

On V , the eigenvalues are $\lambda_1 = x < y = \lambda_2$, with eigenvectors e_1, e_2 . Moreover,

$$\nabla\lambda_1 \cdot e_1 = 1, \quad \nabla\lambda_2 \cdot e_2 = 1.$$

The rank-one-free conclusion follows from [proposition 4.1](#). $\square$

Remark 4.3. The preceding corollary solves only the rank-one-free part of the desired construction. It does not assert that the constitutive set contains a nondegenerate T_N .

5. AN EXACT NONDEGENERATE T_4 COMPATIBLE WITH A STRICTLY CONVEX ENTROPY GRAPH

This section separates the finite T_N algebra from the global realization problem. We first exhibit four exact matrices forming a nondegenerate ordered T_4 whose first columns lie on the graph of the quadratic entropy. We then show that these matrices can be matched at the four states by one entropy-compatible polynomial potential. The missing issue is preserving strict hyperbolicity, genuine nonlinearity, and global rank-one-freeness on a convex state space containing all four states.

Let

$$(5.1) \quad \eta(x, y) = \frac{1}{2}(x^2 + y^2)$$

and choose

$$(5.2) \quad U_1 = (-3, 0), \quad U_2 = (-1, 0), \quad U_3 = (1, 0), \quad U_4 = (3, 0).$$

Set

$$(5.3) \quad A_i := \begin{pmatrix} U_i \\ \eta(U_i) \end{pmatrix} \in \mathbb{R}^3,$$

and define the second columns

$$(5.4) \quad B_1 = \begin{pmatrix} -68/5 \\ 0 \\ 24/5 \end{pmatrix}, \quad B_2 = \begin{pmatrix} -94/15 \\ 0 \\ 44/5 \end{pmatrix},$$

$$B_3 = \begin{pmatrix} 98/15 \\ 0 \\ -28/5 \end{pmatrix}, \quad B_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Finally, let

$$(5.5) \quad X_i := (A_i, B_i) \in \mathbb{R}^{3 \times 2}.$$

Explicitly,

$$(5.6) \quad \begin{aligned} X_1 &= \begin{pmatrix} -3 & -68/5 \\ 0 & 0 \\ 9/2 & 24/5 \end{pmatrix}, & X_2 &= \begin{pmatrix} -1 & -94/15 \\ 0 & 0 \\ 1/2 & 44/5 \end{pmatrix}, \\ X_3 &= \begin{pmatrix} 1 & 98/15 \\ 0 & 0 \\ 1/2 & -28/5 \end{pmatrix}, & X_4 &= \begin{pmatrix} 3 & 0 \\ 0 & 0 \\ 9/2 & 0 \end{pmatrix}. \end{aligned}$$

Proposition 5.1 (Exact finite nondegenerate T_4). *The ordered tuple (X_1, X_2, X_3, X_4) in (5.6) is a nondegenerate ordered T_4 -configuration.*

Proof. Define

$$(5.7) \quad \begin{aligned} c_1 &= \begin{pmatrix} -34/15 \\ 0 \\ 4/5 \end{pmatrix}, & c_2 &= \begin{pmatrix} -2/15 \\ 0 \\ -8/5 \end{pmatrix}, \\ c_3 &= \begin{pmatrix} 14/15 \\ 0 \\ -4/5 \end{pmatrix}, & c_4 &= \begin{pmatrix} 22/15 \\ 0 \\ 8/5 \end{pmatrix}, \end{aligned}$$

and

$$(5.8) \quad s_1 = 3, \quad s_2 = -2, \quad s_3 = 7, \quad s_4 = 0.$$

Let

$$(5.9) \quad C_i := c_i \otimes (1, s_i), \quad \kappa_i := 2, \quad i = 1, \dots, 4.$$

Every C_i is nonzero and rank one. Directly,

$$(5.10) \quad \sum_{i=1}^4 c_i = \begin{pmatrix} (-34 - 2 + 14 + 22)/15 \\ 0 \\ (4 - 8 - 4 + 8)/5 \end{pmatrix} = 0,$$

and

$$(5.11) \quad \sum_{i=1}^4 s_i c_i = \begin{pmatrix} (-102 + 4 + 98)/15 \\ 0 \\ (12 + 16 - 28)/5 \end{pmatrix} = 0.$$

Hence

$$(5.12) \quad \sum_{i=1}^4 C_i = \left(\sum_{i=1}^4 c_i, \sum_{i=1}^4 s_i c_i \right) = 0.$$

Set

$$(5.13) \quad P := \left(\begin{pmatrix} 23/15 \\ 0 \\ 29/10 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right).$$

Substitution gives the four identities

$$(5.14) \quad P + 2C_1 = \begin{pmatrix} -3 & -68/5 \\ 0 & 0 \\ 9/2 & 24/5 \end{pmatrix} = X_1,$$

$$(5.15) \quad P + C_1 + 2C_2 = \begin{pmatrix} -1 & -94/15 \\ 0 & 0 \\ 1/2 & 44/5 \end{pmatrix} = X_2,$$

$$(5.16) \quad P + C_1 + C_2 + 2C_3 = \begin{pmatrix} 1 & 98/15 \\ 0 & 0 \\ 1/2 & -28/5 \end{pmatrix} = X_3,$$

$$(5.17) \quad P + C_1 + C_2 + C_3 + 2C_4 = \begin{pmatrix} 3 & 0 \\ 0 & 0 \\ 9/2 & 0 \end{pmatrix} = X_4.$$

Thus the ordered T_4 relations hold.

Since the second row of every X_i vanishes, the rank of $X_i - X_j$ is two if and only if the determinant obtained from rows one and three is nonzero. The six determinants are

$$(5.18) \quad \begin{array}{c|cccccc} (i, j) & (1, 2) & (1, 3) & (1, 4) & (2, 3) & (2, 4) & (3, 4) \\ \hline \det_{13}(X_i - X_j) & \frac{112}{3} & \frac{584}{15} & -\frac{144}{5} & -\frac{144}{5} & -\frac{904}{15} & \frac{112}{3} \end{array}$$

All are nonzero. Hence $\text{rank}(X_i - X_j) = 2$ for every $i \neq j$. $\square$

5.1. Exact entropy-compatible interpolation of the four matrices. The finite configuration above is more than merely compatible with the entropy graph in its first column: it can be realized at the four selected states by one potential system with the quadratic entropy.

Suppose $F = \nabla \Phi$ and $q = U \cdot \nabla \Phi - \Phi$. Since the second column of $G(U_i)$ is B_i , the interpolation requirements are

$$(5.19) \quad \nabla \Phi(U_i) = \begin{pmatrix} -(B_i)_1 \\ -(B_i)_2 \end{pmatrix}, \quad \Phi(U_i) = (B_i)_3 + U_i \cdot \nabla \Phi(U_i).$$

For the data in (5.2)–(5.4), this gives

$$(5.20) \quad \begin{array}{c|ccc} U_i & \Phi(U_i) & \Phi_x(U_i) & \Phi_y(U_i) \\ \hline (-3, 0) & -36 & 68/5 & 0 \\ (-1, 0) & 38/15 & 94/15 & 0 \\ (1, 0) & -182/15 & -98/15 & 0 \\ (3, 0) & 0 & 0 & 0 \end{array}$$

The unique polynomial ϕ of degree at most seven satisfying the eight one-dimensional Hermite conditions

$$\phi(x_i) = \Phi(U_i), \quad \phi'(x_i) = \Phi_x(U_i), \quad x_i \in \{-3, -1, 1, 3\},$$

is

$$(5.21) \quad \phi(x) = \frac{(x-3)^2}{5760} \left(28x^5 + 75x^4 - 1502x^3 - 7548x^2 - 7718x - 807 \right).$$

Consequently, for arbitrary constants $L, \gamma \in \mathbb{R}$, the polynomial potential

$$(5.22) \quad \Phi_{L,\gamma}(x, y) := \phi(x) + \frac{L}{2}y^2 + \frac{\gamma}{6}y^3$$

satisfies all the jet conditions in (5.20). Therefore, with

$$(5.23) \quad F_{L,\gamma} = \nabla \Phi_{L,\gamma}, \quad q_{L,\gamma} = U \cdot \nabla \Phi_{L,\gamma} - \Phi_{L,\gamma},$$

we have

$$(5.24) \quad G_{F_{L,\gamma}, \eta, q_{L,\gamma}}(U_i) = X_i, \quad i = 1, \dots, 4.$$

This exact interpolation does not solve the desired hyperbolic realization problem. Indeed,

$$(5.25) \quad \phi'''(x) = \frac{1}{240} (245x^4 - 465x^3 - 4250x^2 + 2139x + 6013).$$

Let the numerator be denoted by $R(x)$. Exact evaluation gives

$$(5.26) \quad R(-2) = -7625 < 0 < R(-1) = 334$$

and

$$(5.27) \quad R(1) = 3682 > 0 > R(2) = -6509.$$

Hence ϕ''' vanishes at least once in $(-2, -1)$ and at least once in $(1, 2)$. For the separable system (5.22), the coordinate vector e_1 is an eigenvector of $DF_{L,\gamma}$ and the corresponding genuine-nonlinearity quantity, whenever that eigenvalue is simple, is $\phi'''(x)$. At each zero of ϕ''' , either the two eigenvalues coincide, in which case strict hyperbolicity fails, or the e_1 characteristic field is simple and its genuine-nonlinearity quantity vanishes. Therefore no choice of L and γ makes this separable interpolation simultaneously strictly hyperbolic and genuinely nonlinear on a convex set containing all four states.

There remains substantial coupled interpolation freedom. For example, one may add to (5.22) a term

$$(5.28) \quad \Psi(x, y) := yh(x), \quad h(-3) = h(-1) = h(1) = h(3) = 0,$$

without changing the values or first derivatives of Φ at the four states. Such a term changes the mixed Hessian through $h'(x_i)$ and can rotate the characteristic directions. What is not established here is a choice of the higher-order interpolation freedom for which strict hyperbolicity, genuine nonlinearity, and the absence of every rank-one connection hold throughout one convex neighborhood of the four states.

6. AN EXACT FINITE NONDEGENERATE T_6 COMPATIBLE WITH THE ENTROPY GRAPH

The preceding T_4 shows that strict convexity of the entropy graph is not, by itself, an obstruction to the finite matrix algebra. Since the original question specifically suggested increasing the number of states, we now give an entirely rational nondegenerate ordered T_6 . As in section 5, this is a finite interpolation result, not yet a global rank-one-free hyperbolic realization.

6.1. The six matrices and their ordered parameterization. Retain the quadratic entropy

$$(6.1) \quad \eta(x, y) = \frac{1}{2}(x^2 + y^2)$$

and choose the six states

$$(6.2) \quad \begin{aligned} U_1 &= (-2, 0), & U_2 &= (-1, 1), & U_3 &= (0, -1), \\ U_4 &= (1, 2), & U_5 &= (2, -1), & U_6 &= (3, 1). \end{aligned}$$

Define the following matrices, whose first columns are $(U_i, \eta(U_i))^T$:

$$\begin{aligned}
 (6.3) \quad X_1 &= \begin{pmatrix} -2 & 0 \\ 0 & 0 \\ 2 & 0 \end{pmatrix}, & X_2 &= \begin{pmatrix} -1 & -22/21 \\ 1 & 16/21 \\ 1 & -113/63 \end{pmatrix}, \\
 X_3 &= \begin{pmatrix} 0 & -41/21 \\ -1 & 110/21 \\ 1/2 & 415/126 \end{pmatrix}, & X_4 &= \begin{pmatrix} 1 & 0 \\ 2 & 5 \\ 5/2 & 5/2 \end{pmatrix}, \\
 X_5 &= \begin{pmatrix} 2 & 89/21 \\ -1 & -38/21 \\ 5/2 & 479/126 \end{pmatrix}, & X_6 &= \begin{pmatrix} 3 & -38/21 \\ 1 & -22/21 \\ 5 & -178/63 \end{pmatrix}.
 \end{aligned}$$

Proposition 6.1 (Exact finite nondegenerate T_6). *The ordered tuple $(X_1, \dots, X_6)$ in (6.3) is a nondegenerate ordered T_6 -configuration.*

Proof. Define

$$\begin{aligned}
 (6.4) \quad c_1 &= \begin{pmatrix} -43/21 \\ -5/21 \\ -50/63 \end{pmatrix}, & c_2 &= \begin{pmatrix} -11/21 \\ 8/21 \\ -113/126 \end{pmatrix}, & c_3 &= \begin{pmatrix} 5/21 \\ -17/21 \\ -44/63 \end{pmatrix}, \\
 c_4 &= \begin{pmatrix} 13/21 \\ 23/21 \\ 41/63 \end{pmatrix}, & c_5 &= \begin{pmatrix} 17/21 \\ -20/21 \\ 41/126 \end{pmatrix}, & c_6 &= \begin{pmatrix} 19/21 \\ 11/21 \\ 89/63 \end{pmatrix},
 \end{aligned}$$

and

$$(6.5) \quad (s_1, s_2, s_3, s_4, s_5, s_6) = (0, 1, -3, 1, 3, -2).$$

Set

$$(6.6) \quad C_i := c_i \otimes (1, s_i), \quad \kappa_i := 2, \quad i = 1, \dots, 6.$$

Every C_i is nonzero and has rank one. The two columnwise closure identities are exact. Indeed,

$$(6.7) \quad \sum_{i=1}^6 c_i = \begin{pmatrix} (-43 - 11 + 5 + 13 + 17 + 19)/21 \\ (-5 + 8 - 17 + 23 - 20 + 11)/21 \\ (-100 - 113 - 88 + 82 + 41 + 178)/126 \end{pmatrix} = 0,$$

and

$$(6.8) \quad \sum_{i=1}^6 s_i c_i = \begin{pmatrix} (-11 - 15 + 13 + 51 - 38)/21 \\ (8 + 51 + 23 - 60 - 22)/21 \\ (-113 + 264 + 82 + 123 - 356)/126 \end{pmatrix} = 0.$$

Consequently,

$$(6.9) \quad \sum_{i=1}^6 C_i = \left(\sum_{i=1}^6 c_i, \sum_{i=1}^6 s_i c_i \right) = 0.$$

Take

$$(6.10) \quad P = \begin{pmatrix} 44/21 & 0 \\ 10/21 & 0 \\ 226/63 & 0 \end{pmatrix}.$$

The six ordered outer-state identities are

$$(6.11) \quad P + 2C_1 = X_1,$$

$$(6.12) \quad P + C_1 + 2C_2 = X_2,$$

$$(6.13) \quad P + C_1 + C_2 + 2C_3 = X_3,$$

$$(6.14) \quad P + C_1 + C_2 + C_3 + 2C_4 = X_4,$$

$$(6.15) \quad P + C_1 + C_2 + C_3 + C_4 + 2C_5 = X_5,$$

$$(6.16) \quad P + C_1 + C_2 + C_3 + C_4 + C_5 + 2C_6 = X_6.$$

For an additional transparent check, the polygon vertices $P_i = P + \sum_{j < i} C_j$ have first and second columns p_i and z_i given by

$$(6.17) \quad \begin{array}{c|cc} i & p_i^\top & z_i^\top \\ \hline 1 & (44/21, 10/21, 226/63) & (0, 0, 0) \\ 2 & (1/21, 5/21, 176/63) & (0, 0, 0) \\ 3 & (-10/21, 13/21, 239/126) & (-11/21, 8/21, -113/126) \\ 4 & (-5/21, -4/21, 151/126) & (-26/21, 59/21, 151/126) \\ 5 & (8/21, 19/21, 233/126) & (-13/21, 82/21, 233/126) \\ 6 & (25/21, -1/21, 137/63) & (38/21, 22/21, 178/63) \end{array}$$

Thus $P_i = (p_i, z_i)$ and, row by row,

$$(6.18) \quad X_i = P_i + 2C_i = (p_i + 2c_i, z_i + 2s_i c_i), \quad i = 1, \dots, 6,$$

which is equivalent to (6.11)–(6.16).

It remains to verify nondegeneracy. For every pair, the determinant of the 2×2 submatrix formed by rows one and two of $X_i - X_j$ is nonzero. The fifteen exact values are

$$(6.19) \quad \begin{array}{c|ccccc} (i, j) & (1, 2) & (1, 3) & (1, 4) & (1, 5) & (1, 6) \\ \hline \det_{12}(X_i - X_j) & 38/21 & 179/21 & 15 & -3 & -24/7 \end{array}$$

$$(6.20) \quad \begin{array}{c|ccccc} (i, j) & (2, 3) & (2, 4) & (2, 5) & (2, 6) & (3, 4) \\ \hline \det_{12}(X_i - X_j) & 8/3 & 52/7 & 20/7 & -152/21 & -128/21 \end{array}$$

$$(6.21) \quad \begin{array}{c|ccccc} (i, j) & (3, 5) & (3, 6) & (4, 5) & (4, 6) & (5, 6) \\ \hline \det_{12}(X_i - X_j) & -296/21 & -134/7 & 124/21 & -292/21 & 90/7 \end{array}$$

Hence every difference $X_i - X_j$ has rank at least two. Since it is a 3×2 matrix, its rank is exactly two. Therefore the six matrices are pairwise rank-two connected, and the ordered T_6 is nondegenerate. $\square$

6.2. Exact interpolation by one entropy-compatible potential. The six matrices can also be realized exactly at their six states by one polynomial potential system. Write the second column of X_i as B_i and set

$$(6.22) \quad g_i := \begin{pmatrix} -(B_i)_1 \\ -(B_i)_2 \end{pmatrix}, \quad \varphi_i := (B_i)_3 + U_i \cdot g_i.$$

The required value and gradient data are

i	U_i	φ_i	$(g_i)_1$	$(g_i)_2$
1	$(-2, 0)$	0	0	0
2	$(-1, 1)$	$-227/63$	$22/21$	$-16/21$
3	$(0, -1)$	$1075/126$	$41/21$	$-110/21$
4	$(1, 2)$	$-15/2$	0	-5
5	$(2, -1)$	$-817/126$	$-89/21$	$38/21$
6	$(3, 1)$	$230/63$	$38/21$	$22/21$

Indeed, for a potential flux $F = \nabla \Phi$ and $q = U \cdot \nabla \Phi - \Phi$, the identities

$$(6.24) \quad \Phi(U_i) = \varphi_i, \quad \nabla \Phi(U_i) = g_i$$

are exactly equivalent to $G(U_i) = X_i$.

An explicit interpolating potential can be chosen in the form

$$(6.25) \quad \Phi_6(x, y) := a(x) + b(x)y,$$

where

$$(6.26) \quad b(x) := -\frac{(x+2)(33x^4 + 49x^3 - 623x^2 + 141x + 1100)}{420}$$

and

$$(6.27) \quad a(x) := -\frac{(x+2)^2}{18144000} (782221x^9 - 7916652x^8 + 27685989x^7 \\ - 29453268x^6 - 38283441x^5 + 98693292x^4 \\ - 39490369x^3 - 23127372x^2 + 21009600x - 14940000).$$

We record the interpolation mechanism. If $U_i = (x_i, y_i)$, then b is the unique polynomial of degree at most five satisfying

$$(6.28) \quad b(x_i) = (g_i)_2, \quad i = 1, \dots, 6.$$

Once b is fixed, a is the unique polynomial of degree at most eleven satisfying the twelve Hermite conditions

$$(6.29) \quad a(x_i) = \varphi_i - y_i b(x_i), \\ a'(x_i) = (g_i)_1 - y_i b'(x_i), \quad i = 1, \dots, 6.$$

The polynomials in (6.26)–(6.27) are the resulting exact interpolants. Consequently,

$$(6.30) \quad \Phi_6(U_i) = \varphi_i, \quad \nabla \Phi_6(U_i) = g_i, \quad i = 1, \dots, 6.$$

Define

$$(6.31) \quad F_6 := \nabla \Phi_6, \quad q_6(U) := U \cdot \nabla \Phi_6(U) - \Phi_6(U).$$

By lemma 2.2, (η, q_6) is an entropy–entropy-flux pair for F_6 and $D^2 \eta = I_2$. Equations (6.24) and (6.30) yield

$$(6.32) \quad G_{F_6, \eta, q_6}(U_i) = X_i, \quad i = 1, \dots, 6.$$

Thus the polynomial constitutive surface contains the exact nondegenerate ordered T_6 at the six displayed states.

Remark 6.2 (What this finite T_6 does and does not prove). The construction proves that increasing from four to six states creates ample exact finite-dimensional freedom even after one imposes a common strictly convex entropy graph and the entropy compatibility relation at all six states. It does not prove that F_6 is strictly hyperbolic and genuinely nonlinear on a convex set containing the six states, nor that the entire surface K_{F_6, η, q_6} is rank-one-free there. Those are global inequalities and two-point conditions, respectively; finite Hermite interpolation alone does not preserve them.

7. A POSITIVE-WEIGHT DETERMINANT IDENTITY FOR EVERY ORDERED T_N

We now derive a necessary condition valid for every $N \geq 4$. It depends only on the outer matrices and positive weights determined by the ordered parameterization.

For $1 \leq r < s \leq 3$, let

$$(7.1) \quad \pi_{rs} : \mathbb{R}^{3 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$$

be the projection onto rows r and s , and write

$$(7.2) \quad \det_{rs} X := \det(\pi_{rs} X).$$

Theorem 7.1 (Weighted minor identity). *Let $(X_1, \dots, X_N)$ be a weak ordered T_N -configuration with parameters P, C_i, κ_i . Define α_i, β_i by (2.16) and*

$$(7.3) \quad \theta_i := \frac{\alpha_i \prod_{j=i+1}^N \beta_j}{1 - \prod_{j=1}^N \beta_j}, \quad i = 1, \dots, N,$$

where an empty product equals one. Then

$$(7.4) \quad \theta_i > 0, \quad \sum_{i=1}^N \theta_i = 1,$$

and, for every $1 \leq r < s \leq 3$,

$$(7.5) \quad \boxed{\sum_{1 \leq i < j \leq N} \theta_i \theta_j \det_{rs}(X_i - X_j) = 0.}$$

Proof. Iterating the convex recursion (2.17) from P_1 to $P_{N+1} = P_1$ gives

$$(7.6) \quad P_1 = \left(\prod_{j=1}^N \beta_j \right) P_1 + \sum_{i=1}^N \alpha_i \left(\prod_{j=i+1}^N \beta_j \right) X_i.$$

Since $0 < \beta_j < 1$, the denominator in (7.3) is positive. Rearranging (7.6) yields

$$(7.7) \quad P_1 = \sum_{i=1}^N \theta_i X_i.$$

The coefficients are positive. Applying the same identity to the constant scalar component of an affine space, or simply comparing the coefficients in the iterated affine recursion, gives $\sum_i \theta_i = 1$. More explicitly,

$$\sum_{i=1}^N \alpha_i \prod_{j=i+1}^N \beta_j = \sum_{i=1}^N (1 - \beta_i) \prod_{j=i+1}^N \beta_j$$

$$= 1 - \prod_{j=1}^N \beta_j,$$

which proves (7.4).

Fix a row pair (r, s) and set

$$A_i := \pi_{rs} P_i, \quad D_i := \pi_{rs} C_i, \quad Y_i := \pi_{rs} X_i.$$

Because C_i has rank one, D_i has rank at most one, and therefore

$$(7.8) \quad \det(A_i + tD_i) = \det A_i + t \operatorname{cof}(A_i) : D_i$$

contains no quadratic term in t . In particular, determinant is affine on the segment joining A_i and $Y_i = A_i + \kappa_i D_i$. Since $A_{i+1} = \beta_i A_i + \alpha_i Y_i$, we obtain

$$(7.9) \quad \det A_{i+1} = \beta_i \det A_i + \alpha_i \det Y_i.$$

Iterating (7.9) around the cycle exactly as above gives

$$(7.10) \quad \det \pi_{rs} P_1 = \sum_{i=1}^N \theta_i \det \pi_{rs} X_i.$$

On the other hand, applying π_{rs} to (7.7) gives

$$\pi_{rs} P_1 = \sum_{i=1}^N \theta_i \pi_{rs} X_i.$$

For arbitrary 2×2 matrices Y_i and weights summing to one, the quadratic identity

$$(7.11) \quad \sum_{i=1}^N \theta_i \det Y_i - \det \left(\sum_{i=1}^N \theta_i Y_i \right) = \sum_{1 \leq i < j \leq N} \theta_i \theta_j \det(Y_i - Y_j)$$

holds. To verify it, polarize the quadratic form $Y \mapsto \det Y$, or expand both sides entry by entry. The left-hand side of (7.11) vanishes by (7.10), and (7.5) follows. $\square$

Corollary 7.2 (Same-sign obstruction). *Fix a row pair (r, s) . If*

$$(7.12) \quad \det_{rs}(X_i - X_j) > 0 \quad \text{for every } i < j,$$

or if all these determinants are strictly negative, then $(X_1, \dots, X_N)$ cannot be a weak ordered T_N -configuration.

Proof. Every coefficient $\theta_i \theta_j$ in (7.5) is strictly positive, so a positive weighted sum of numbers having one strict sign cannot vanish. $\square$

Remark 7.3 (Checks on the finite T_4 and T_6). For the parameterization in proposition 5.1, all $\kappa_i = 2$, so $\alpha_i = \beta_i = 1/2$. Formula (7.3) gives

$$(7.13) \quad (\theta_1, \theta_2, \theta_3, \theta_4) = \frac{1}{15}(1, 2, 4, 8).$$

Using the six determinants in (5.18), one checks exactly that

$$\sum_{1 \leq i < j \leq 4} \theta_i \theta_j \det_{13}(X_i - X_j) = 0.$$

Thus the signs in (5.18) necessarily change.

Likewise, all parameters of the T_6 in proposition 6.1 equal two, and hence

$$(7.14) \quad (\theta_1, \dots, \theta_6) = \frac{1}{63}(1, 2, 4, 8, 16, 32).$$

Exact calculation gives both the barycenter identity

$$(7.15) \quad \sum_{i=1}^6 \theta_i X_i = P$$

and, for each of the three row pairs,

$$(7.16) \quad \sum_{1 \leq i < j \leq 6} \theta_i \theta_j \det_{rs}(X_i - X_j) = 0, \quad (r, s) \in \{(1, 2), (1, 3), (2, 3)\}.$$

The mixed signs visible already in (6.21) are therefore forced by the ordered T_6 geometry.

8. SEPARABLE RANK-ONE-FREE SYSTEMS AND A ONE-DIMENSIONAL OBSTRUCTION

The next class gives a direct proof of rank-one-freeness without requiring a cubic potential and simultaneously shows why placing six states on one coordinate line cannot work.

Let $I, J \subset \mathbb{R}$ be open intervals and let $\varphi \in C^3(I)$, $\psi \in C^3(J)$. Define on the rectangle $V := I \times J$

$$(8.1) \quad \Phi(x, y) := \varphi(x) + \psi(y),$$

$$(8.2) \quad F(x, y) = (\varphi'(x), \psi'(y)), \quad \eta(x, y) = \frac{1}{2}(x^2 + y^2),$$

and

$$(8.3) \quad q(x, y) = x\varphi'(x) - \varphi(x) + y\psi'(y) - \psi(y).$$

Proposition 8.1 (A separable rank-one-free class). *Assume*

$$(8.4) \quad \varphi'''(x) \neq 0 \quad \text{for every } x \in I, \quad \psi'''(y) \neq 0 \quad \text{for every } y \in J,$$

and assume that the ranges of φ'' and ψ'' are disjoint; for example,

$$(8.5) \quad \sup_{x \in I} \varphi''(x) < \inf_{y \in J} \psi''(y).$$

Then the system is strictly hyperbolic and genuinely nonlinear on V , and $K_{F, \eta, q}$ has no rank-one connections.

Proof. The Jacobian is diagonal:

$$(8.6) \quad DF(x, y) = \text{diag}(\varphi''(x), \psi''(y)).$$

The range separation gives strict hyperbolicity. The eigenvectors are e_1, e_2 , and the corresponding genuine-nonlinearity quantities are $\varphi'''(x)$ and $\psi'''(y)$, which are nonzero by (8.4).

Suppose that $U = (x, y)$ and $W = (\bar{x}, \bar{y})$ are distinct and satisfy the rank-one conditions of lemma 2.4 for some σ . Set

$$a := \bar{x} - x, \quad b := \bar{y} - y.$$

The Rankine–Hugoniot equations are

$$(8.7) \quad \varphi'(\bar{x}) - \varphi'(x) = \sigma a, \quad \psi'(\bar{y}) - \psi'(y) = \sigma b.$$

If $a \neq 0$ and $b \neq 0$, the mean value theorem gives points ξ between x and $\bar{x}$ and ζ between y and $\bar{y}$ such that

$$(8.8) \quad \sigma = \varphi''(\xi) = \psi''(\zeta),$$

contradicting the disjointness of the two ranges.

Suppose next that $a \neq 0$ and $b = 0$. The entropy equality is then

$$(8.9) \quad \begin{aligned} & \bar{x}\varphi'(\bar{x}) - \varphi(\bar{x}) - x\varphi'(x) + \varphi(x) \\ &= \sigma \frac{\bar{x}^2 - x^2}{2}. \end{aligned}$$

Using the first identity in (8.7) to eliminate σ , we obtain

$$(8.10) \quad \frac{\bar{x} - x}{2} (\varphi'(x) + \varphi'(\bar{x})) - (\varphi(\bar{x}) - \varphi(x)) = 0.$$

Since φ''' is continuous and never vanishes on the interval I , it has a fixed sign. Therefore φ' is strictly convex or strictly concave. The strict trapezoidal inequality says that the left-hand side of (8.10) is strictly positive in the convex case and strictly negative in the concave case, after orienting the interval from the smaller to the larger endpoint. It cannot vanish. The case $a = 0, b \neq 0$ is identical with ψ in place of φ . Hence no distinct pair can satisfy the rank-one conditions. $\square$

We now apply the determinant identity to any finite collection of states on a horizontal line. Fix $y_0 \in J$. In the row pair (1, 3), the constants depending only on y_0 disappear from differences, so the relevant constitutive curve is

$$(8.11) \quad Y(x) := \begin{pmatrix} x & -\varphi'(x) \\ x^2/2 & \varphi(x) - x\varphi'(x) \end{pmatrix}.$$

Lemma 8.2 (Sign of the projected pairwise determinants). *For $a < b$ in I ,*

$$(8.12) \quad \det(Y(b) - Y(a)) = (b - a) \left[\varphi(b) - \varphi(a) - \frac{b - a}{2} (\varphi'(a) + \varphi'(b)) \right].$$

If $\varphi''' > 0$ on I , this determinant is strictly negative for every $a \neq b$; if $\varphi''' < 0$, it is strictly positive for every $a \neq b$.

Proof. Subtract the two matrices in (8.11) and expand the determinant:

$$\begin{aligned} \det(Y(b) - Y(a)) &= (b - a) [\varphi(b) - b\varphi'(b) - \varphi(a) + a\varphi'(a)] \\ &\quad + \frac{b^2 - a^2}{2} [\varphi'(b) - \varphi'(a)]. \end{aligned}$$

Using $b^2 - a^2 = (b - a)(a + b)$ and collecting the endpoint derivative terms gives (8.12).

If $\varphi''' > 0$, then φ' is strictly convex, so

$$(8.13) \quad \int_a^b \varphi'(x) \, dx < \frac{b - a}{2} (\varphi'(a) + \varphi'(b)).$$

Since the integral is $\varphi(b) - \varphi(a)$, the bracket in (8.12) is negative. The concave case has the opposite sign. Finally, replacing $Y(b) - Y(a)$ by its negative does not change a 2×2 determinant, so the sign is independent of the ordering of the pair. $\square$

Theorem 8.3 (No one-dimensional T_N in the separable class). *Under the assumptions of proposition 8.1, no tuple of distinct states*

$$(8.14) \quad U_i = (x_i, y_0), \quad i = 1, \dots, N,$$

can produce an ordered T_N -configuration in $K_{F,\eta,q}$, for any $N \geq 4$. The analogous statement holds for states on a vertical line.

Proof. By lemma 8.2, all nonzero determinants

$$\det_{13}(G(U_i) - G(U_j))$$

afford the same strict sign. This contradicts corollary 7.2. The vertical-line case uses rows (2, 3) and ψ in place of φ . $\square$

Remark 8.4 (The special cubic-in- y model). Taking

$$\psi(y) = \frac{L}{2}y^2 + \frac{\gamma}{6}y^3, \quad \gamma \neq 0,$$

gives

$$\psi''(y) = L + \gamma y, \quad \psi'''(y) = \gamma.$$

If the range of $L + \gamma y$ on J is disjoint from the range of φ'' on I , all conclusions of this section apply. In particular, placing six candidate states on $y = 0$ cannot produce a T_6 .

9. A REGULAR SMALL-AMPLITUDE OBSTRUCTION

We now show that increasing N does not resolve the local obstruction near a strictly hyperbolic state. The conclusion is not that every sufficiently small exact T_N is impossible. Rather, any one-scale family that collapses regularly to a point must be collinear at first order; a genuinely two-dimensional small configuration would have to be singular or multiscale.

9.1. Normal form at a base state.

Lemma 9.1 (Entropy-variable normal form). *Let $U_0 \in V$. After an invertible affine change of state variables and an invertible affine transformation of the constitutive matrices, one may assume*

$$(9.1) \quad U_0 = 0, \quad F(0) = 0, \quad \eta(0) = q(0) = 0, \quad \nabla \eta(0) = \nabla q(0) = 0,$$

$$(9.2) \quad D^2\eta(0) = I_2, \quad DF(0) = A := \text{diag}(\lambda_1, \lambda_2), \quad \lambda_1 < \lambda_2.$$

These transformations preserve rank-one connections and ordered T_N -configurations.

Proof. First translate the state so that $U_0 = 0$ and subtract the constant $F(U_0)$ from the flux. Let

$$H := D^2\eta(U_0) > 0, \quad A_0 := DF(U_0).$$

By lemma 2.1, HA_0 is symmetric. Let $S := H^{1/2}$. Then

$$SA_0S^{-1}$$

is symmetric, because $HA_0 = A_0^T H$. Choose an orthogonal matrix O such that

$$O^T SA_0 S^{-1} O = \text{diag}(\lambda_1, \lambda_2).$$

Set

$$(9.3) \quad W := T(U - U_0), \quad T := O^T S,$$

and define

$$\tilde{F}(W) := T(F(U_0 + T^{-1}W) - F(U_0)).$$

Then $D\tilde{F}(0) = \text{diag}(\lambda_1, \lambda_2)$.

To remove the affine part of the entropy, define

$$(9.4) \quad \tilde{\eta}(W) := \eta(U) - \eta(U_0) - \nabla \eta(U_0) \cdot (U - U_0),$$

$$(9.5) \quad \tilde{q}(W) := q(U) - q(U_0) - \nabla \eta(U_0) \cdot (F(U) - F(U_0)),$$

where $U = U_0 + T^{-1}W$. A direct chain-rule calculation gives

$$\nabla_W \tilde{q} = \nabla_W \tilde{\eta} D_W \tilde{F}.$$

Moreover,

$$D_W^2 \tilde{\eta}(0) = T^{-\top} H T^{-1} = I_2.$$

At the matrix level, the new constitutive map is obtained from the old one by subtracting $G(U_0)$ and applying an invertible row operation: the top two rows are multiplied by T , and the affine entropy row is replaced by the old entropy row minus $\nabla \eta(U_0)$ times the top two rows. Thus the transformation has the form $X \mapsto L(X - G(U_0))$ with $L \in GL(3)$, and it preserves all relevant ranks and T_N relations. $\square$

In the normal form,

$$(9.6) \quad F(U) = AU + O(|U|^2),$$

$$(9.7) \quad \eta(U) = \frac{1}{2}|U|^2 + O(|U|^3),$$

$$(9.8) \quad q(U) = \frac{1}{2}U^\top AU + O(|U|^3).$$

The last formula follows because $D^2 q(0) = D^2 \eta(0) D F(0) = A$ in the normalized coordinates.

9.2. Regular one-scale collapsing families. We now state the precise asymptotic hypothesis.

Definition 9.2 (Regular one-scale collapsing family). Let $\varepsilon \downarrow 0$. For each ε , let

$$X_i^\varepsilon = G(U_i^\varepsilon), \quad i = 1, \dots, N,$$

form a weak ordered T_N with parameters $P^\varepsilon, C_i^\varepsilon, \kappa_i^\varepsilon$. Let

$$(9.9) \quad P_i^\varepsilon := P^\varepsilon + \sum_{j < i} C_j^\varepsilon$$

be the polygon vertices. The family is *regular one-scale at U_0* if, after the normalization of [lemma 9.1](#) and after passing to a subsequence,

$$(9.10) \quad U_i^\varepsilon = \varepsilon \hat{U}_i + o(\varepsilon),$$

$$(9.11) \quad \kappa_i^\varepsilon \longrightarrow \kappa_i \in (1, \infty),$$

and the first-column state parts of the polygon edges satisfy

$$(9.12) \quad \frac{1}{\varepsilon} (C_i^\varepsilon)_{\{1,2\},1} \longrightarrow c_i \neq 0.$$

The bounded nondegenerate limits in (9.11) exclude the singular regimes $\kappa_i \downarrow 1$ and $\kappa_i \rightarrow \infty$. Condition (9.12) says that every rank-one arm is visible at the same first-order state scale.

Theorem 9.3 (First-order collinearity). *Let $F, \eta, q \in C^3$ near a strictly hyperbolic state U_0 , with $D^2 \eta(U_0) > 0$. For every regular one-scale collapsing family of weak ordered T_N -configurations, all first-order state-space arms c_i belong to the same characteristic eigenspace of $DF(U_0)$. Consequently, the first-order polygon and all first-order outer states are collinear.*

Proof. Work in the normal form of [lemma 9.1](#). Write

$$A = \text{diag}(\lambda_1, \lambda_2), \quad \lambda_1 < \lambda_2.$$

We split the proof into four steps.

Step 1: first- and second-order limits of the polygon vertices. By (2.17), each polygon vertex P_i^ε is a convex combination of the outer matrices X_j^ε , with positive coefficients depending only on

the κ_j^ε . Since (9.11) keeps these parameters in a compact subset of $(1, \infty)$, the coefficients stay bounded. The expansions (9.6)–(9.8) and (9.10) imply

$$(9.13) \quad (X_i^\varepsilon)_{\{1,2\},\{1,2\}} = O(\varepsilon), \quad (X_i^\varepsilon)_{3,\{1,2\}} = O(\varepsilon^2).$$

The same bounds hold for P_i^ε and for their differences $C_i^\varepsilon = P_{i+1}^\varepsilon - P_i^\varepsilon$.

After passing to a further subsequence, write

$$(9.14) \quad \frac{1}{\varepsilon}(P_i^\varepsilon)_{\{1,2\},\{1,2\}} \longrightarrow (p_i, v_i) \in \mathbb{R}^{2 \times 2},$$

$$(9.15) \quad \frac{1}{\varepsilon}(C_i^\varepsilon)_{\{1,2\},\{1,2\}} \longrightarrow (c_i, d_i),$$

and

$$(9.16) \quad \frac{(P_i^\varepsilon)_{31}}{\varepsilon^2} \longrightarrow h_i, \quad \frac{(P_i^\varepsilon)_{32}}{\varepsilon^2} \longrightarrow k_i.$$

The polygon and outer-state relations give

$$(9.17) \quad p_{i+1} = p_i + c_i, \quad v_{i+1} = v_i + d_i,$$

$$(9.18) \quad \widehat{U}_i = p_i + \kappa_i c_i, \quad -A\widehat{U}_i = v_i + \kappa_i d_i.$$

Step 2: every first-order arm is a characteristic direction. Define

$$(9.19) \quad w_i := v_i + A p_i.$$

Using (9.18), we find

$$(9.20) \quad w_i = -\kappa_i(Ac_i + d_i).$$

Using (9.17),

$$(9.21) \quad \begin{aligned} w_{i+1} &= v_i + d_i + A(p_i + c_i) = w_i + (d_i + Ac_i) \\ &= \left(1 - \frac{1}{\kappa_i}\right) w_i = \beta_i w_i. \end{aligned}$$

After one circuit,

$$(9.22) \quad w_1 = \left(\prod_{i=1}^N \beta_i\right) w_1.$$

Since every $\beta_i \in (0, 1)$, the product is strictly less than one. Therefore

$$(9.23) \quad w_i = 0 \quad \text{for every } i.$$

It follows from (9.20) that

$$(9.24) \quad d_i = -Ac_i.$$

Because each C_i^ε has rank one, the limiting top block (c_i, d_i) has rank at most one. Since $c_i \neq 0$, equations (9.24) and

$$\text{rank}(c_i, -Ac_i) \leq 1$$

imply that Ac_i is parallel to c_i . Strict hyperbolicity then gives an index $\ell_i \in \{1, 2\}$ such that

$$(9.25) \quad Ac_i = \lambda_{\ell_i} c_i.$$

Step 3: positivity of the entropy defect. Set

$$(9.26) \quad e_i := h_{i+1} - h_i.$$

The first-column third-row component of $X_i^\varepsilon = P_i^\varepsilon + \kappa_i^\varepsilon C_i^\varepsilon$, together with (9.7), yields

$$(9.27) \quad \frac{1}{2} |\widehat{U}_i|^2 = h_i + \kappa_i e_i.$$

Define the defect of the polygon vertex from the quadratic entropy graph by

$$(9.28) \quad \delta_i := h_i - \frac{1}{2} |p_i|^2.$$

Using $p_{i+1} = p_i + c_i$, $\widehat{U}_i = p_i + \kappa_i c_i$, and (9.27), we obtain

$$(9.29) \quad \begin{aligned} \delta_{i+1} &= h_i + e_i - \frac{1}{2} |p_i + c_i|^2 \\ &= \left(1 - \frac{1}{\kappa_i}\right) \delta_i + \frac{\kappa_i - 1}{2} |c_i|^2. \end{aligned}$$

For clarity, the algebra is

$$\begin{aligned} h_i + e_i &= \left(1 - \frac{1}{\kappa_i}\right) h_i + \frac{1}{2\kappa_i} |p_i + \kappa_i c_i|^2 \\ &= \left(1 - \frac{1}{\kappa_i}\right) h_i + \frac{1}{2\kappa_i} |p_i|^2 + p_i \cdot c_i + \frac{\kappa_i}{2} |c_i|^2, \end{aligned}$$

and subtracting $\frac{1}{2} |p_i + c_i|^2$ gives (9.29).

Writing

$$(9.30) \quad a_i := \frac{\kappa_i - 1}{2} |c_i|^2 > 0,$$

iteration around the cycle gives

$$(9.31) \quad \delta_1 = \frac{\sum_{i=1}^N a_i \prod_{j=i+1}^N \beta_j}{1 - \prod_{j=1}^N \beta_j} > 0.$$

Then (9.29) implies

$$(9.32) \quad \delta_i > 0, \quad i = 1, \dots, N.$$

Step 4: comparison of the entropy-flux defect across successive arms. Because the first-column top part of C_i^ε is nonzero for small ε , rank one implies that there is a scalar τ_i^ε such that

$$(9.33) \quad (C_i^\varepsilon)_{\cdot,2} = \tau_i^\varepsilon (C_i^\varepsilon)_{\cdot,1}.$$

By (9.24) and (9.25),

$$(9.34) \quad \tau_i^\varepsilon \longrightarrow -\lambda_{\ell_i}.$$

Taking the third row in (9.33) at order ε^2 gives

$$(9.35) \quad k_{i+1} - k_i = -\lambda_{\ell_i} (h_{i+1} - h_i) = -\lambda_{\ell_i} e_i.$$

Define the entropy-flux defect of the vertex by

$$(9.36) \quad \zeta_i := k_i + \frac{1}{2} p_i^\top A p_i.$$

The second-column third-row component of the outer-state identity, together with (9.8), gives

$$(9.37) \quad -\frac{1}{2} \widehat{U}_i^\top A \widehat{U}_i = k_i + \kappa_i (k_{i+1} - k_i).$$

Using (9.35), $\widehat{U}_i = p_i + \kappa_i c_i$, and $Ac_i = \lambda_{\ell_i} c_i$, equation (9.37) becomes

$$-\frac{1}{2}(p_i + \kappa_i c_i)^\top A(p_i + \kappa_i c_i) = k_i - \kappa_i \lambda_{\ell_i} e_i.$$

Using (9.27) to write $\kappa_i e_i = \frac{1}{2} |\widehat{U}_i|^2 - h_i$, all mixed and c_i -quadratic terms cancel, leaving

$$(9.38) \quad \zeta_i = -\lambda_{\ell_i} \delta_i.$$

For completeness, this cancellation is

$$\begin{aligned} k_i &= -\frac{1}{2} \widehat{U}_i^\top A \widehat{U}_i + \lambda_{\ell_i} \left(\frac{1}{2} |\widehat{U}_i|^2 - h_i \right) \\ &= -\frac{1}{2} p_i^\top A p_i - \lambda_{\ell_i} \left(h_i - \frac{1}{2} |p_i|^2 \right), \end{aligned}$$

which is exactly (9.38).

The same arm also transports the defect to the next polygon vertex. Indeed, using (9.35), $p_{i+1} = p_i + c_i$, and $Ac_i = \lambda_{\ell_i} c_i$, we get

$$\begin{aligned} \zeta_{i+1} - \zeta_i &= -\lambda_{\ell_i} e_i + p_i^\top Ac_i + \frac{1}{2} c_i^\top Ac_i \\ &= -\lambda_{\ell_i} \left(e_i - p_i \cdot c_i - \frac{1}{2} |c_i|^2 \right) \\ &= -\lambda_{\ell_i} (\delta_{i+1} - \delta_i). \end{aligned}$$

Together with (9.38), this yields

$$(9.39) \quad \zeta_{i+1} = -\lambda_{\ell_i} \delta_{i+1}.$$

Applying (9.38) to the next arm gives instead

$$(9.40) \quad \zeta_{i+1} = -\lambda_{\ell_{i+1}} \delta_{i+1}.$$

Since $\delta_{i+1} > 0$, comparison of (9.39) and (9.40) gives

$$(9.41) \quad \lambda_{\ell_i} = \lambda_{\ell_{i+1}}.$$

By induction, every arm belongs to the same characteristic family. Since each eigenspace of A is one-dimensional, all c_i are collinear. The relations $p_{i+1} = p_i + c_i$ and $\widehat{U}_i = p_i + \kappa_i c_i$ then show that both the first-order polygon and the first-order outer states lie on one affine line. $\square$

Corollary 9.4 (No regular genuinely two-dimensional small T_6). *A family of T_6 -configurations collapsing to a strictly hyperbolic state cannot have two linearly independent first-order state-space arms while the parameters κ_i remain uniformly away from 1 and infinity and all arms occur at one common scale.*

Remark 9.5. The theorem leaves open several singular possibilities: some arms may be smaller than others; some κ_i may converge to 1 or diverge; or the first-order polygon may be collinear while a two-dimensional configuration appears only at a higher order. Thus the result is an obstruction to a regular perturbative construction, not a full local nonexistence theorem.

10. THE REMAINING RANK-ONE-FREE T_N PROBLEM

The discussion can now be summarized without ambiguity.

Theorem 10.1 (What is proved in this note). *The following statements hold.*

- (i) *The assertion that strict hyperbolicity, genuine nonlinearity, and one strictly convex entropy rule out weak ordered T_N -configurations is false; indeed, the explicit system in [section 3](#) contains such a configuration for every even $N \geq 4$.*
- (ii) *There are explicit systems satisfying all of the same local differential assumptions whose constitutive sets have no rank-one connections; see [corollary 4.2](#).*
- (iii) *Strict entropy convexity and the finite algebra of nondegenerate ordered T_4 and T_6 configurations are compatible; see [propositions 5.1](#) and [6.1](#). Each displayed tuple can be matched by one entropy-compatible polynomial potential. The separable T_4 realization fails genuine nonlinearity inside the convex hull of its states, and no global hyperbolicity, genuine-nonlinearity, or rank-one-free claim is made for the T_6 interpolant.*
- (iv) *Every ordered T_N satisfies the weighted determinant identities [\(7.5\)](#).*
- (v) *No horizontal or vertical one-dimensional construction works in the separable rank-one-free class of [proposition 8.1](#), for any N .*
- (vi) *Every regular one-scale small-amplitude family is first-order collinear, independently of N .*

What is *not* proved is the following existence statement.

Problem 10.2 (Rank-one-free constitutive T_N). Construct an integer $N \geq 4$, a nonempty convex open set $V \subset \mathbb{R}^2$, a flux $F: V \rightarrow \mathbb{R}^2$, and an entropy pair (η, q) such that

- (i) $D^2\eta > 0$ on V ;
- (ii) DF is strictly hyperbolic on V ;
- (iii) both characteristic fields are genuinely nonlinear on V ;
- (iv) $K_{F,\eta,q}$ has no rank-one connections;
- (v) $\mathcal{K}_{F,\eta,q}^{(N)}$ contains a nondegenerate ordered T_N -configuration.

The preceding results show that a solution of [problem 10.2](#), if it exists, cannot be obtained by the most immediate constructions. In particular, a candidate should satisfy all of the following checks.

- (a) It must be genuinely two-dimensional at finite amplitude, or else use a singular multiscale mechanism invisible at first order.
- (b) For each row pair (r, s) , its projected pairwise determinants must satisfy the positive-weight cancellation [\(7.5\)](#); one strict sign is impossible.
- (c) All strict inequalities—entropy convexity, spectral separation, and genuine nonlinearity—must hold on the entire convex hull of the selected states, because every convex admissible state space containing those states also contains their convex hull.
- (d) Rank-one-freeness must be checked globally on that convex state space, not merely for the finitely many outer states.

The stronger question in which the system is additionally assumed to satisfy the global Liu entropy condition was posed explicitly in [\[KS24\]](#); the authors conjectured that no T_4 exists in that regime. That question does not settle [problem 10.2](#): the absence of rank-one connections is weaker than the global Liu condition, and allowing $N = 6$ introduces configurations not covered by a T_4 theorem. On the other hand, the results of [\[LP20, JT24, PV24\]](#) show that even for the p -system there are substantial nonexistence mechanisms extending beyond rank-one exclusion alone.

APPENDIX A. EXACT SYMBOLIC VERIFICATION

The source package accompanying this note contains the script `verify_TN_computations.py`. It performs all calculations in exact rational arithmetic using SymPy. In particular, it verifies:

- (i) the derivatives of the potential in [\(3.4\)](#);

- (ii) the resultant identity (3.26);
- (iii) the Sturm endpoint sign table (3.29);
- (iv) the four exact T_4 identities (5.14)–(5.17);
- (v) the six pairwise determinants in (5.18);
- (vi) the Hermite polynomial (5.21);
- (vii) the six exact T_6 identities (6.11)–(6.16);
- (viii) all fifteen pairwise rank-two checks for the finite T_6 ;
- (ix) the explicit T_6 potential interpolation (6.26)–(6.32);
- (x) the weighted cancellations for both finite configurations and all three row pairs of the T_6 .

The following compact excerpt reproduces the resultant and Sturm checks.

```
import sympy as sp

x, t = sp.symbols("x t")
Q = (sp.Rational(7, 6)*(3*x**2 - 1)*t**2
      + (9*x**2 + 5*x - 2)*t
      - sp.Rational(7, 6)*(3*x**2 - 1))
H = 18*x + 21*x*t - 15*t**2 - (20 + 25*x)*t**3

P = (129007296*x**8 + 235355760*x**7 + 80940141*x**6
      - 64110420*x**5 - 27496701*x**4 + 10403280*x**3
      + 1618495*x**2 - 1203500*x + 236425)

assert sp.factor(sp.resultant(Q, H, t) - P/216) == 0

sturm = sp.sturm(P, x)
signs_minus = [sp.sign(s.subs(x, -1)) for s in sturm]
signs_plus = [sp.sign(s.subs(x, 1)) for s in sturm]
print(signs_minus)
print(signs_plus)
```

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